Práctica dos: Sistema cardiovascular

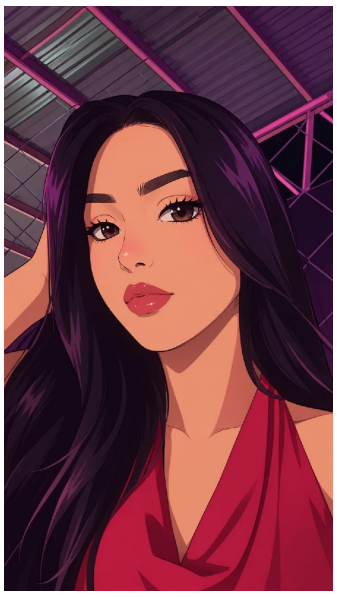
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

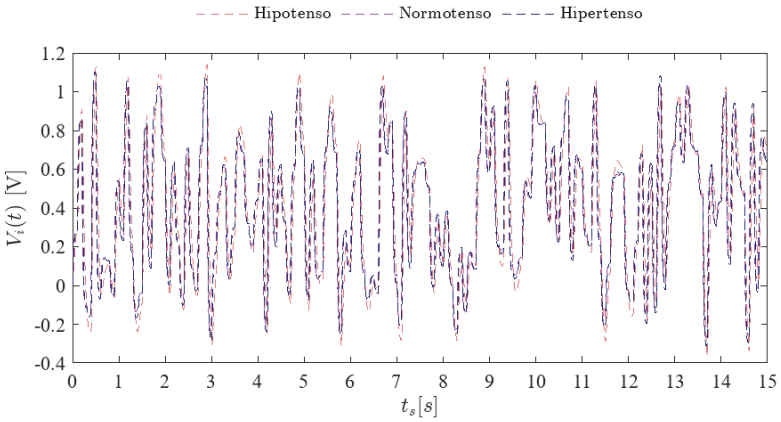
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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('Practica')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x = sim('Practica',parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(Pa,'signal.xlsx');
```

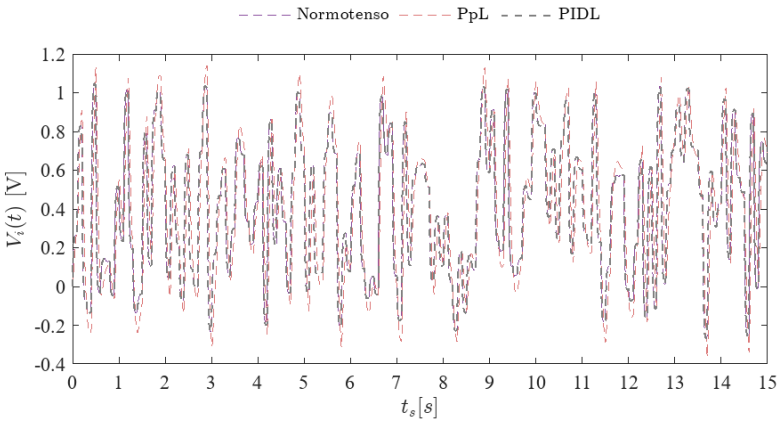
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH, 'Individuos')
```



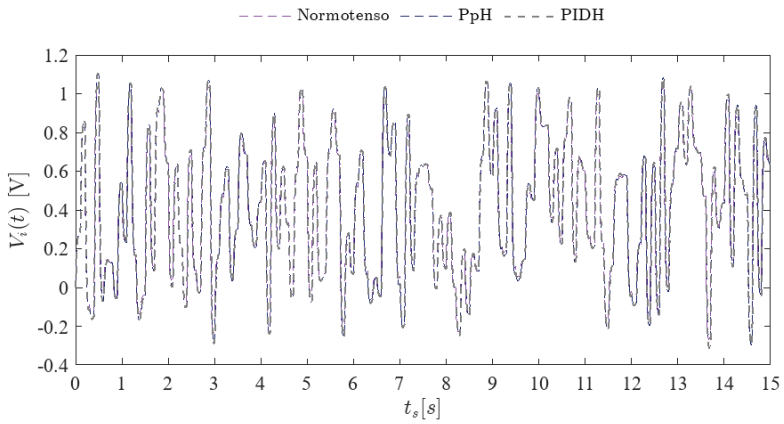
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t, x.PpN, x.PpL, x.PIDL, 'Hipotenso')
```



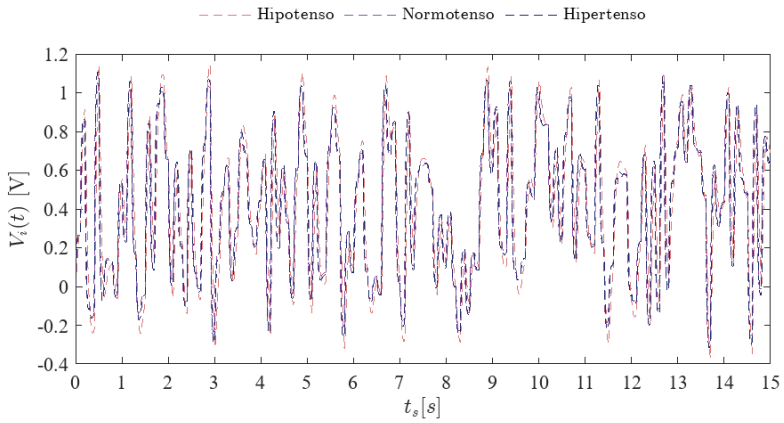
Respuestas: Normotenso, Hipertenso y Controlador

```
plotsignals(x.t, x.PpN, x.PpH, x.PIDH, 'Hipertenso')
```



Modelo de EDOs

```
par_hipo  = [0.01,0.001,1.00,3.500]; %[Z,L,R,C]
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'ode')
```




```
function Pp = cardio(par, Pa, t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;

    for i = 1:n
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
```

```
end
```

Repuesta del sistema

```
function plotsignals(t, x1, x2, x3, caso)

    set(figure(), 'color', 'w')
    set(gcf, 'units', 'inches', 'position', [1, 1, 7, 3.5])
    set(gca, 'fontname', 'times new roman', 'fontsize', 11)
    hold on; grid off; box on;

    clr1 = [220, 120, 120]/255;
    clr2 = [140, 80, 160]/255;
    clr3 = [30, 40, 110]/255;
    clr4 = [0.5, 0.5, 0.5];


    if caso == "Individuos"
        plot(t,x1,'--', 'LineWidth',0.25, 'Color',clr1)
        plot(t,x2,'--', 'LineWidth',0.25, 'Color',clr2)
        plot(t,x3,'--', 'LineWidth',0.25, 'Color',clr3)
        L = legend('Hipotenso', 'Normotenso', 'Hipertenso');
    elseif caso == "Hipotenso"
        plot(t,x1,'--', 'LineWidth',0.25, 'Color',clr2)
        plot(t,x2,'--', 'LineWidth',0.25, 'Color',clr1)
        plot(t,x3,'--', 'LineWidth',1, 'Color',clr4)
        L = legend('Normotenso', 'PpL', 'PIDL');
    elseif caso == "Hipertenso"
        plot(t,x1,'--', 'LineWidth',0.25, 'Color',clr2)
        plot(t,x2,'--', 'LineWidth',0.25, 'Color',clr3)
        plot(t,x3,'--', 'LineWidth',1, 'Color',clr4)
        L = legend('Normotenso', 'PpH', 'PIDH');
    elseif caso == "ode"
        plot(t,x1,'--', 'LineWidth',0.25, 'Color',clr1)
        plot(t,x2,'--', 'LineWidth',0.25, 'Color',clr2)
        plot(t,x3,'--', 'LineWidth',0.25, 'Color',clr3)
        L = legend('Hipotenso', 'Normotenso', 'Hipertenso');
    end


set(L,'interpreter','latex','location','northoutside','orientation','horizontal',
'box','off')
    xlabel('$t_s[s]$','interpreter','latex')
    ylabel('$V_i(t)$ [V]','interpreter','latex')
    xlim([0 15]); xticks(0:1:15)
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
```

```
    exportgraphics(gcf, [caso, 'matlab.pdf'], 'contenttype', 'vector')
end
```